$$\tau = \mu \frac{dV}{dr} \qquad V_c = \frac{(\Delta p + \rho g \Delta z) D^2}{16 \mu L} \qquad V_{avg} = \frac{1}{2} V_c$$

$$Re = \frac{\rho V_{avg} D}{\mu}$$

For laminar flow,

$Re \leq 2100$

For turbulent flow,

$Re > 4000$

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L$$

$$h_L = f \frac{L}{D} \frac{V_{avg}^2}{2g}$$

Velocity: LT^{-1}

Acceleration: LT^{-2}

Density: ML^{-3}

Force: MLT^{-2}

Pressure: $ML^{-1}T^{-2}$

Viscosity: $ML^{-1}T^{-1}$

Torque: $ML^{2}T^{-2}$

Power: $ML^{2}T^{-3}$

Rotation speed: T^{-1}

Volumetric flow rate: $L^{3}T^{-1}$